

Suggested Design Project Work Sequence

1. Show the class how we arrived at the [Yachats Off Grid 2024.csv](https://nsrdb.nlr.gov/data-viewer) data set.
<https://nsrdb.nlr.gov/data-viewer>
`Gen_Yachats_Off_Grid_2024.ipynb`
2. Show the class how to fetch data files and other materials in general from
<https://osu-fifem.github.io/OSU-ESE450/>
3. Students build plotting tools in Python to explore the load data set.
4. Work on a first iteration of the system design. What do you think is the best way to satisfy the requirements? Calculate rough sizing and do preliminary (rough) cost analysis.
5. Simulation – I will help you when we get to this. You will write a control law function and a state update function.
 - a. The **control law** will determine the power setpoints and generation levels for each controllable device in the system and output the controls as a vector.
e.g. The heating system will be thermostatically controlled, so the control law for the heating system might look something like:
$$\text{Heater Power} = UA * (T_{\text{indoor}} - T_{\text{outdoor}})$$
 - b. The **state update** function will update the state of any devices and environmental conditions such as house temperature given their previous state and given the control vector. e.g.
Updated Battery Energy: $E_{\text{batt},n+1} = E_{\text{batt},n} + P_{\text{batt},\text{charge}} \cdot \Delta t$
6. Visualization of simulation results.
7. Economic analysis – Using the simulation results, calculate the actual cost of operating the system for the 2024 calendar year.
8. What-Ifs. What if something fails? How resilient is your system?
9. Iterate on design and repeat simulation, visualization, and economic analysis.